

DATA COLLECTION AND MANAGEMENT STRATEGY

Supporting Document for a Virtual Clinical Study

Virtual RCT of a Smartphone-Delivered CBT Application for Mild-to-Moderate Generalized Anxiety in Adults

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Week 2 Task: Data Collection and Management Strategy

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1. Introduction and Purpose

This document is a companion to the Week 1 Clinical Study Protocol (VCR-2026-001) and describes the data collection and management strategy for the virtual randomized controlled trial (RCT) evaluating a smartphone-delivered CBT application for adults with mild-to-moderate generalized anxiety. The study enrolls 120 participants (60 per arm) across a 12-week participation window per subject (8-week intervention plus a 4-week follow-up), with outcome data captured at four fixed time points: baseline, Week 4, Week 8, and Week 12.

Because the study is fully decentralized, no data is collected on paper or in person; every data point originates from an electronic source. At the study's planned scale, this generates an estimated $120 \text{ participants} \times 4 \text{ time points} \times \sim 48 \text{ structured data fields per time point} \approx 23,040$ discrete data points from questionnaires alone, plus continuous passive engagement logs from the CBT application (estimated 15–20 events per participant per week). A clearly defined data strategy is therefore essential to protecting data integrity, participant privacy, and analytical validity at this volume.

The purpose of this strategy is to define exactly what data will be collected, from where, using which named tools and versions, how it will be stored and secured with specific technical controls, how it will be cleaned and validated using explicit, numeric rules, and who is responsible for each step on a defined timeline. Where a specific tool, threshold, or approach was chosen, a brief justification is provided so the reasoning behind each decision is traceable.

2. Key Data Elements

The table below identifies the core data elements required to answer the study's research question and support safety monitoring, organized by data domain, with the specific coding/range used for each element.

Domain	Data Element (field name)	Type / Coding	Collection Point
Eligibility	age_yrs; gad7_prescreen; device_os_ver	Integer (18–65); Integer (0–21); String	Screening
Demographics	sex; education_lvl; prior_therapy_yn	Categorical (M/F/Other); Ordinal (1–5); Boolean	Baseline
Consent	econsent_ts; comprehension_score	ISO 8601 timestamp; Integer (0–5, pass ≥ 4)	Enrollment
Primary Outcome	gad7_total (7 items, each 0–3)	Integer, range 0–21	Baseline, Wk 4, 8, 12
Secondary Outcome	phq9_total (0–27); sus_score (0–100)	Integer; Float (0.5 increments)	Baseline, Wk 4, 8, 12
Engagement	login_ct; modules_complete_ct (0–8); session_min	Integer; Integer; Float (minutes)	Continuous (passive)
Safety	phq9_item9_flag; ae_severity (1–5)	Boolean; Ordinal	Continuous

Domain	Data Element (field name)	Type / Coding	Collection Point
Administrative	study_id (format VCR-###); arm (A/B)	String; Categorical	Throughout study

Example of a single de-identified record row, illustrating how these fields populate in practice:

study_id	arm	time point	gad7_total	phq9_total	session_min
VCR-014	A (CBT app)	Week 4	11	8	142.5
VCR-014	A (CBT app)	Week 8	7	5	168.0
VCR-057	B (waitlist)	Week 8	13	10	0.0

Justification: fields are limited to those directly required to test the primary hypothesis (GAD-7 change), the pre-specified secondary hypotheses (PHQ-9, usability), and safety monitoring (item 9 suicide-risk flag), following the data-minimization principle in HIPAA and GDPR — collecting only what is analytically necessary reduces both privacy risk and the burden of cleaning unused fields.

3. Data Collection Plan

3.1 Data Sources

- Self-report questionnaires (GAD-7, PHQ-9, SUS) completed by participants through the study web portal at 4 fixed time points (Baseline, Week 4, Week 8, Week 12).
- Passive usage data captured automatically by the CBT application's analytics SDK (estimated 15–20 timestamped events per participant per week: session start, module complete, session end).
- Study staff logs from two virtual check-in calls per participant (Week 1, Week 8), entered directly into the EDC within 24 hours of each call.
- Screening and demographic data entered by participants during the online intake process (estimated 8–10 minute form, 22 fields).

3.2 Collection Tools and Instruments

All questionnaire data will be collected using REDCap 14.x (Research Electronic Data Capture), a widely validated, HIPAA-compliant EDC platform, configured with electronic versions of the GAD-7 (Spitzer et al., 2006), PHQ-9, and the 10-item System Usability Scale (Brooke, 1996). Each instrument uses REDCap's built-in branching logic and required-field validation (e.g., an item response outside 0–3 is rejected before the form can be submitted) to reduce missing or inconsistent entries at the point of collection. The CBT application logs engagement events to a Postgres-based analytics database that syncs to the EDC system nightly at 02:00 UTC via a secure REST API (TLS 1.2+), keeping the two data stores within 24 hours of consistency at all times.

Justification: REDCap was selected over a general-purpose survey tool (e.g., Google Forms) because it provides an audit trail (21 CFR Part 11-aligned), built-in role-based permissions, and native support for longitudinal/repeating instruments — all requirements for a study collecting 4 repeated timepoints under IRB oversight.

3.3 Use of Simulated and Publicly Available Data

Since this is a hypothetical/training exercise, no real participant data exists yet. To design, test, and validate the data pipeline described in this document before real enrollment begins, the study team will use:

- A simulated dataset of n=150 synthetic records (125% of target enrollment, to allow attrition-scenario testing), generated in Python (numpy/pandas) by sampling GAD-7 scores from a normal distribution truncated to 5–14 at baseline ($\mu=9.2$, $\sigma=2.1$, based on published GAD-7 norms) and applying a simulated 20% dropout by Week 12, consistent with the attrition rate assumed in the Week 1 power calculation.
- Publicly available, de-identified anxiety-treatment datasets referenced to benchmark realistic score distributions and engagement patterns, for example the NIMH Data Archive (NDA) and the PHQ/GAD normative data published in Kroenke et al. (2010, Gen Hosp Psychiatry).

No real or identifiable participant data will be used at any stage of this exercise. Justification: simulating data before go-live lets the team pressure-test validation rules (Section 5) and confirm the EDC-to-analytics integration works correctly, without any risk of exposing real participant information during development.

4. Data Storage and Management

4.1 Storage Infrastructure

All electronic data will be stored in a HIPAA-compliant, cloud-hosted EDC environment (AWS us-east-1 primary region) with audit-trail logging enabled on every create/read/update/delete event. Data is partitioned into a clinical database (questionnaire responses, consent records — estimated <50 MB of structured data at full enrollment) and a separate analytics database (app usage logs — estimated 200–300 MB across the study given ~20 events/participant/week over 20 weeks), linked only through the coded study_id.

Justification: separating clinical and behavioral-analytics data into two databases limits the blast radius of any single breach and mirrors the principle of purpose-limitation — engagement analytics staff never need access to raw questionnaire responses, and vice versa.

4.2 Access Control

Role	# of Users	Access Level
Principal Investigator	1	Read-only, all data, de-identified view
Study Coordinator	2	Read/write, consent + check-in log only
Data Manager	1	Read/write, all data, including query resolution
Biostatistician	1	Read-only, locked/frozen datasets only

Multi-factor authentication (MFA) is required for all EDC and analytics-platform logins. Access and export logs are reviewed monthly by the Data Manager, and any export of >10 records at once triggers an automatic email alert to the PI. Justification: capping standard roles at the minimum access needed (least-privilege principle) and flagging bulk exports specifically targets the most common real-world cause of research data breaches — unmonitored bulk downloads.

4.3 Security Measures

- Data encrypted in transit using TLS 1.2 or higher, and at rest using AES-256 — the encryption standard required under the HIPAA Security Rule (45 CFR §164.312(a)(2)(iv)) for electronic protected health information.
- Participant identity is separated from study data via a coded study_id (format VCR-###); the identity-linking key is stored in a physically and logically separate, restricted-access system accessible only to the Study Coordinator.
- Automatic session timeout after 15 minutes of inactivity, and IP-allowlisted admin access on the EDC platform.
- An independent third-party security audit is scheduled annually, with the first audit completed prior to first-participant-enrolled.
- Data retention: identifiable data retained for 7 years post-study-closure per ICH-GCP E6(R2) recordkeeping requirements, then securely deleted using a documented destruction certificate.

5. Data Cleaning, Validation, and Integration

Data quality is maintained through point-of-entry validation, automated post-hoc checks with explicit numeric thresholds, and scheduled manual review:

1. Point-of-entry validation: required fields, hard range checks (e.g., GAD-7 item scores restricted to 0–3, giving a valid total of 0–21), and skip-logic are built directly into the REDCap forms, preventing invalid values from ever being saved.
2. Automated queries: a nightly script flags (a) any visit >7 days outside its ± 3 -day protocol window, (b) any record with >20% missing items on a single instrument, and (c) logical inconsistencies, such as a Week 8 record with a timestamp earlier than the corresponding Week 4 record.
3. Missingness threshold: if a participant's cumulative item-level missingness across all instruments exceeds 10%, the record is flagged for Data Manager outreach to the Study Coordinator within 3 business days, per the missing-data handling approach specified in the Week 1 statistical analysis plan (multiple imputation for MAR data).
4. Manual review: the Data Manager performs a biweekly review of all open queries (target: <5 business days average time-to-resolution) and resolves discrepancies jointly with the Study Coordinator, documenting the resolution rationale in the audit trail.
5. Integration: engagement data from the analytics database is merged with questionnaire data from the EDC nightly at 02:00 UTC using study_id as the linking key; an automated row-count reconciliation check confirms the merged record count matches the source EDC record count exactly, with any mismatch >0 records triggering a same-day alert to the Data Manager.
6. Lock and freeze: once cleaning is complete for a given timepoint (target: within 10 business days of the timepoint's data-collection window closing), that dataset is locked; any post-lock change requires a written justification and PI sign-off, logged with a timestamp and reason code.

6. Backup and Disaster Recovery Plan

- Automated daily incremental backups (retained 30 days) and weekly full backups (retained 12 months) of both the EDC and analytics databases.

- Backups stored in a geographically separate cloud region (AWS us-west-2) from the primary database (us-east-1), so a single-region outage cannot affect both copies.
- Backup restoration is tested quarterly against a staging environment to confirm recoverability, with results logged in the disaster-recovery runbook.
- Defined objectives: Recovery Point Objective (RPO) of 24 hours (maximum acceptable data loss, matching the nightly backup cadence) and Recovery Time Objective (RTO) of 24 hours (maximum acceptable downtime before full restoration).

Justification: a 24-hour RPO/RTO was selected because the study's most frequent input (passive engagement logs) is non-safety-critical and can tolerate a 1-day recovery window, while safety-relevant data (the PHQ-9 item-9 suicide-risk flag) is additionally routed through a separate, real-time alerting pathway that does not depend on the nightly batch pipeline, so a backup-restoration delay would never delay a safety response.

7. Timeline and Responsibilities

Activity	Timeline	Responsible Role
EDC build and validation-rule configuration	Weeks 1–2	Data Manager
Simulated (n=150) data testing of collection pipeline	Weeks 2–3	Data Manager / Biostatistician
Security audit and access-control sign-off	Week 3	Principal Investigator
Data collection tools finalized and locked	Week 3 (end)	Principal Investigator
Ongoing data collection and passive logging	Weeks 4–24 (study conduct)	Study Coordinator / Participants
Biweekly query review and cleaning (<5-day SLA)	Continuous during conduct	Data Manager
Nightly data integration, 02:00 UTC	Continuous	Automated system / IT support
Interim data-quality review	Week 12	Data Manager / PI
Quarterly backup-restoration test	Weeks 8, 20	IT Support
Final data lock and freeze	Week 24	Data Manager / Biostatistician

8. References

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